Gaurav Anand

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Professional Summary

VLSI Design undergraduate at Thapar Institute of Engineering & Technology with a CGPA of 8.17/10 and hands-on experience designing and verifying Verilog systems. Familiar with RTL design, functional verification, synthesis, linting, formal equivalence, and FPGA prototyping using Synopsys, Cadence, and Xilinx tools. Project work includes precision-adaptive arithmetic, a Sobel edge-detection accelerator, and an AMBA-AHB controller.

Education

Thapar Institute of Engineering & Technology Patiala, India
Bachelor of Technology in VLSI Design *Aug 2023 – Present*
CGPA: 8.17/10
Relevant Coursework: Digital System Design, Embedded Systems, Computer Architecture, CMOS VLSI Design

Technical Proficiency

- **Languages:** Verilog, SystemVerilog, TCL, Python, MATLAB
- **EDA & Verification Tools:** Synopsys (VCS, Verdi, Design Compiler, SpyGlass); Cadence (Genus, Conformal LEC, Modus, Xcelium); Xilinx Vivado
- **Protocols:** AMBA-AHB, AXI, UART, JTAG
- **Architecture & Simulation:** Gem5, RISC-V GNU Toolchain, Spike

Key Projects

PA-MAC: Low-Power Precision-Adaptive INT8/BF16 Dot-Product Engine Jul 2026 – Present
Tools: Verilog RTL, UPF, Cadence design flows, Xilinx Kintex-7 KC705, GPDK045 45nm

- Designing a precision-adaptive INT8/BF16 dot-product engine using a shared multiplier architecture to reduce hardware redundancy across integer and floating-point computations.
- Developing a pipelined RTL architecture comprising 4 processing elements and 16 MAC units, with mode-selectable INT8/BF16 datapaths.
- Developing a UPF-based fine-grained power-gating architecture with dedicated power domains for control, core computation, BF16-specific logic, and accumulation.
- Targeting FPGA prototyping on the Xilinx Kintex-7 KC705 and ASIC implementation using GPDK045 45nm technology, with Cadence-based RTL verification and digital design flows.

Systolic Array Accelerator for Real-Time Sobel Edge Detection Jun 2026 – Jul 2026
Tools: VCS, Verdi, Genus, Conformal LEC, Modus, Spike, Vivado, MATLAB

- Designed a streaming systolic-array Sobel accelerator in Verilog with parallel G_x/G_y computation, verified using VCS and Verdi.
- Synthesized using Cadence Genus at 180nm, achieving 32,376 cell area, 44,328 total area, and approximately 7.02mW power.
- Performed RTL-to-gate Conformal LEC and scan-cell insertion using Cadence Modus.
- Evaluated RISC-V workload execution using Spike and deployed on Zybo-7000 for a 640×480 grayscale image, reconstructed in MATLAB.

AMBA-AHB Protocol Controller Design & Verification Feb 2026 – Mar 2026
Tools: VCS, Verdi, SpyGlass, Design Compiler

- Designed an AMBA-AHB bus architecture with Master, Slave, Decoder, and Multiplexer modules supporting single, INCR4, and WRAP4 burst transactions, verified using VCS and Verdi.
- Performed SpyGlass RTL linting and analyzed QoR, timing, area, power, and hierarchy reports.
- Achieved 100MHz timing closure with zero setup/hold violations.
- Integrated $4 \times 256 \times 32$ SRAM macros, reducing standard cell count from 95K to approximately 7K and power from 172mW to 30–40mW.

Achievements and Honors

50-Day RTL Design Challenge Jun 2026 – Present
GitHub: gauravanand10

- Completing a 50-day RTL design challenge by designing and verifying one Verilog problem daily.

Additional Information

- **Vocational Training:** Patiala Locomotive Works, Jun 2025 – Jul 2025; tested integrated circuits using the Advance Diagnostic Tester.
- **Technical Interests:** RTL design and verification, low-power VLSI, computer architecture, and RISC-V systems.